

Exponent Laws with Variable BasesQuestion Bank 3.T1 **ANSWER KEY**

Show your work. Circle your final answer.

For questions 1 through 6, write an equivalent expression as a single monomial. Assume no variable equals 0.

1. $x^4 \cdot x^9$
 x^{13}

2. $(n^5)^6$
 n^{30}

3. $(3y^4)^3$
 $27y^{12}$

The 3 is cubed as well.

4. $\frac{a^9}{a^3}$
 a^6

5. $(-6x^2)(2x^7)$
 $-12x^9$

6. $\frac{20p^5q^3}{5p^2q}$
 $4p^3q^2$

Coefficients divide; each variable's exponents subtract.

For questions 7 and 8, apply more than one law. Write each answer as a single monomial.

7. Write an equivalent expression.

a. $(2x^3y)(5x^2y^4)$

$$10x^5y^5$$

b. $(4m^2n^3)^2$

$$16m^4n^6$$

c. $(a^2b)^3 \cdot ab^4$

$$a^7b^7$$

$$a^6b^3 \cdot ab^4.$$

8. Write an equivalent expression.

a. $\frac{(3m^2n)^3}{9m^4n}$

$$3m^2n^2$$

$$27m^6n^3 \text{ over } 9m^4n.$$

b. $\frac{x^8y^5}{x^8y^2}$

$$y^3$$

$$x^0 = 1.$$

c. $\frac{(2k^4)^3}{4k^7}$

$$2k^5$$

$$8k^{12} \text{ over } 4k^7.$$

Working backwards.

9. Find the value of n that makes each statement true.

a. $x^3 \cdot x^n = x^{11}$

$$n = 8$$

b. $(y^n)^4 = y^{20}$

$$n = 5$$

c. $\frac{m^9}{m^n} = m^2$

$$n = 7$$

$$9 - n = 2.$$

10. A rectangle has length $6x^4$ and width $2x^3$.

a. Write its area as a single monomial.

$$12x^7$$

length $\times$ width; coefficients multiply, exponents add.

b. A square has side length $4y^5$. Write its area as a single monomial.

$$16y^{10}$$

$(4y^5)^2$ — the 4 is squared too.

11. Which expression is equivalent to $(5m^3)^2$?

☐ A. $5m^6$

☐ B. $25m^5$

☐ C. $10m^6$

☒ D. $25m^6$

D

12. Select ALL of the expressions that are equivalent to x^{12} .

☒ A. $(x^4)^3$

☐ B. $x^2 \cdot x^6$

☒ C. $(x^6)^2$

☒ D. $\frac{x^{15}}{x^3}$

☐ E. $x^4 \cdot x^3$

☒ F. $x^4 \cdot x^8$

A, C, D and F